

AUTOSAR Architecture Report

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Architecture Summary

The AUTOSAR architecture layers the system into microcontroller, ECU, services, RTE, and application layers, each providing a distinct abstraction. The microcontroller abstraction transforms hardware-specific drivers into a uniform interface, while the ECU abstraction decouples external device access from physical layout. Together, these layers enable software components to be developed, tested, and migrated independently of the underlying hardware.

Dependency Summary

The application software relies on the RTE for communication, which in turn depends on the services layer for OS, memory, and diagnostics. The RTE is generated from ECU-specific configurations that depend on the ECU abstraction layer, which itself uses the microcontroller abstraction to access hardware drivers. Complex drivers are integrated through the microcontroller and ECU layers, ensuring timing-critical functions are available to the RTE and services.

Summary Statistics

Metric	Count
Total Components	10
Interfaces	7
Ports	6
Messages	7
Functional Flows	5

Functional Flows

Microcontroller Abstraction Layer Workflow

Purpose: Transforms microcontroller-specific internal drivers into a standardized interface, enabling higher software layers to remain independent of the underlying microcontroller.

ECU Abstraction Layer Workflow

Purpose: Provides a hardware-independent API for external devices, decoupling higher software layers from the ECU's physical hardware layout.

Complex Drivers Integration Workflow

Purpose: Enables integration of special-purpose or non-AUTOSAR-specified drivers that require high timing constraints or migration support, bridging hardware to the RTE.

Services Layer Workflow

Purpose: Provides core operating system, communication, memory, diagnostic, and state-management services to applications, RTE, and other basic software modules.

AUTOSAR Runtime Environment (RTE) Workflow

Purpose: Generates ECU-specific RTE instances that provide communication services to application software, making software components independent of ECU mapping.

Extracted Entities

Name	Type	Page	Context
Crypto Service Manager	Component	4	...tial network support for CAN/LIN slave. ■New Ethernet stack extens
J1939 modules	Component	4	...ervice Manager to System Services ■Revised presentation of J1939
Output Compare Unit Drive	Component	4	...ement concepts: “Pretended Networking”, “ECU Degradation” ■Add
Time Service	Component	4	...king”, “ECU Degradation” ■Added new modules: “Output Compare
FlexRay Transport Layer	Component	4	...11-12-22 4.0.3 AUTOSAR Administration ■Added a note for the R
Ethernet stack extensions	Component	4	...inistration ■Clarification of partial network support for CAN/LIN slave
CAN	Interface	3	...ted new 4.2 concepts for: Switch Configuration; Sender-Receiver-Se
LIN	Interface	4	...1.1 AUTOSAR Administration ■Clarification of partial network supp
Ethernet	Interface	4	...inistration ■Clarification of partial network support for CAN/LIN slave
FlexRay	Interface	4	...11-12-22 4.0.3 AUTOSAR Administration ■Added a note for the R
J1939	Interface	4	...ons ■Added Crypto Service Manager to System Services ■Revised
CAN port	Port	4	Extracted Port
LIN port	Port	4	Extracted Port
Ethernet port	Port	4	Extracted Port
FlexRay port	Port	4	Extracted Port
J1939 messages	Message	4	Extracted Network Message
FlexRay messages	Message	4	Extracted Network Message
CAN messages	Message	4	Extracted Network Message
LIN messages	Message	4	Extracted Network Message
Ethernet frames	Message	50	...Geo-Networking = Layer 3 (Addressing based on geographic areas,
Energy Management	Component	4	...ervices ■Revised presentation of J1939 and added new J1939 mod

Diagnostic Event Manager (DEM)	Component	133	...uction code) Each application software component (SW-C) can rep
EcuM	Component	4	...ion of term AUTOSAR-ECU on slide "94jt1" ■Corrected CDD acces
System Basis Chip	Component	4	...2009-12-18 4.0.1 AUTOSAR Administration ■Added a note regard
DEM symbol generation interface	Interface	4	Extracted Interface
CDD access interface	Interface	4	Extracted Interface
AUTOSAR-ECU port	Port	4	Extracted Port
System Basis Chip port	Port	4	Extracted Port
DBG text	Signal	4	Extracted Signal
DLT text	Signal	4	...ding support for System Basis Chips on slide "94juq" ■Clarification
AUTOSAR-ECU	ECU	4	...g DEM symbol generation ■Fixed minor typography issues ■Clarifi